

Econ 2 - Lecture 9 - 2/9/26

Discussion Activity #3 → Can you forecast the future of the economy? Can AI?

Lecture Quiz #5 Released Wednesday (2/11), Due 2/18

⇒ No class on Monday, 2/16

This week: All about equilibrium! $Y = AE$!

Last Class:

↓ Production → Spending

Consumption (C) is 70% of Aggregate Expenditures (AE)

What determines C?

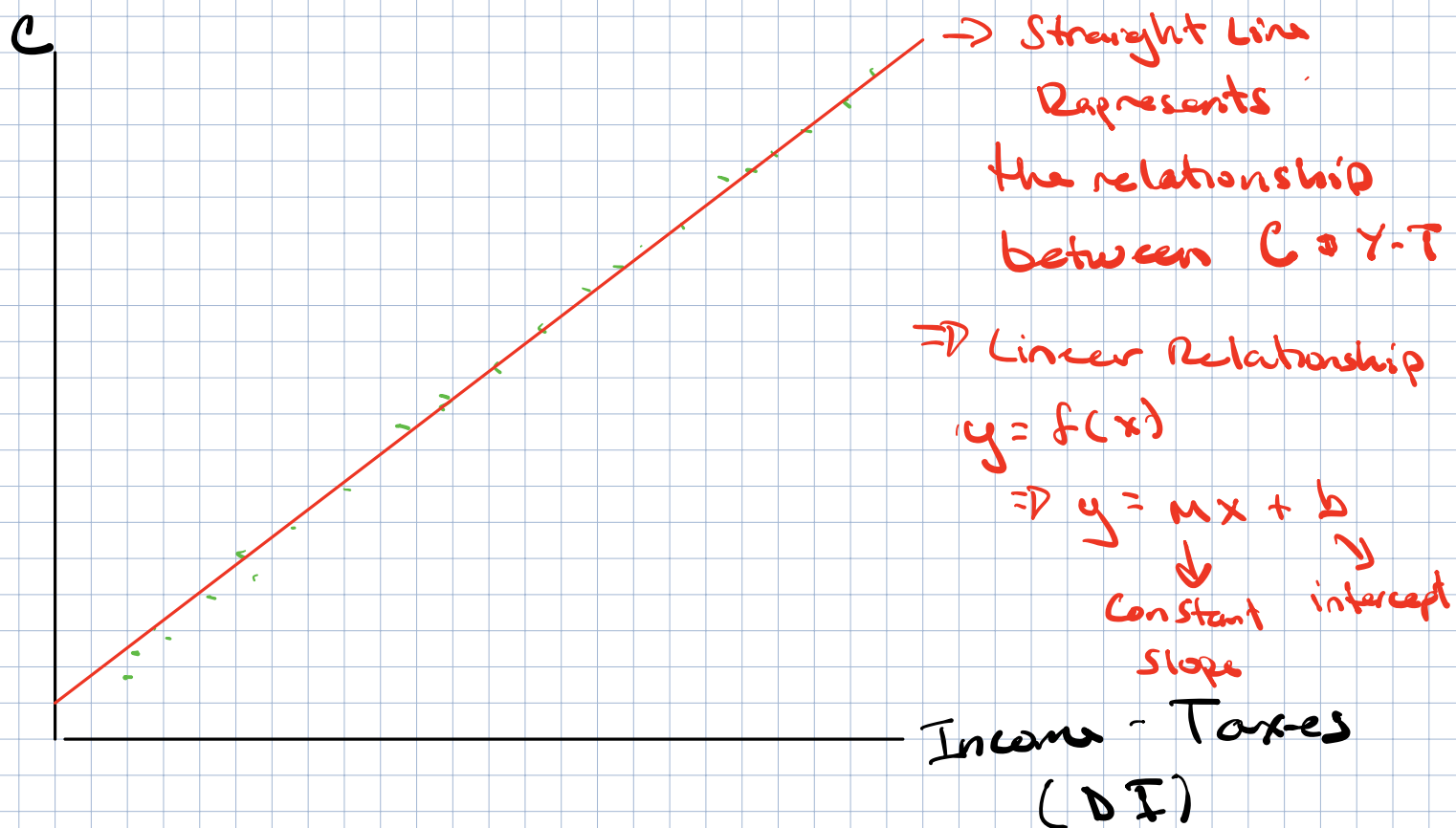
- 1.) Income (Y), Disposable Income ($DI = Y - T$)
- 2.) Wealth = Assets - Liabilities
- 3.) Interest Rates (r): Cost of borrowing / Reward for saving
Related to prices (P) → Week 9

As $r \uparrow$, $C \downarrow$, As $r \downarrow$, $C \uparrow$

- 4.) Expectations: What does future Y look like?
- 5.) Preferences: Kids, family size, lockdown

How important is Y (income) to C (consumption)?

How important is income (Y) to consumption (C)?



Convert general linear function into Consumption function

$$C = \text{Intercept} + \text{Slope} (Y - T)$$

Intercept?

Value of outcome when $Y - T = 0$

Level of C when $Y - T = 0$

→ food, water, utilities, healthcare, necessities

→ Baseline level of spending ⇒ Win lottery? Wealth ↑ regardless of $Y - T$

→ Source of funding if $Y - T = 0$? Savings, borrowing, Gov't, family, friends (?), etc.

Value of Investments

Injection of cash
↑ Intercept

$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\text{Change in } C}{\text{Change in } Y-T} \Rightarrow \frac{\Delta C}{\Delta Y-T}$$

Constant Slope = Linear $\rightarrow$ Straight Line

Same increase in C for every additional take-home dollar increase ($\Delta Y \uparrow * 1$)

Fraction of $Y-T$ we put towards C

Consumption function

C = Autonomous
Consumption
(AC)

$\downarrow$
Word cloud
from lecture 8

+ Marginal
Propensity
to Consume
(MPC)

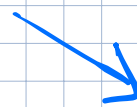


Additional
Take-home Dollar

Two options



Spend
MPC



Save

Marginal
Propensity
to Save
(MPS)

$$MPC + MPS = 1$$

$$C = AC + MPC(Y-T)$$

$$C = AC + mpc (Y - T)$$

$Y - T$	C
0	2000
2000	3200
4000	4400
6000	5600
8000	6800
10000	8000

$AC = \$2000$
 when $Y - T = 0$
 $C = 2000$

What is the mpc?

Increase $Y - T$ by \$2000
 Increase C by \$1200

Increase $Y - T$ by \$1000 $\Rightarrow$ Increase C by \$600

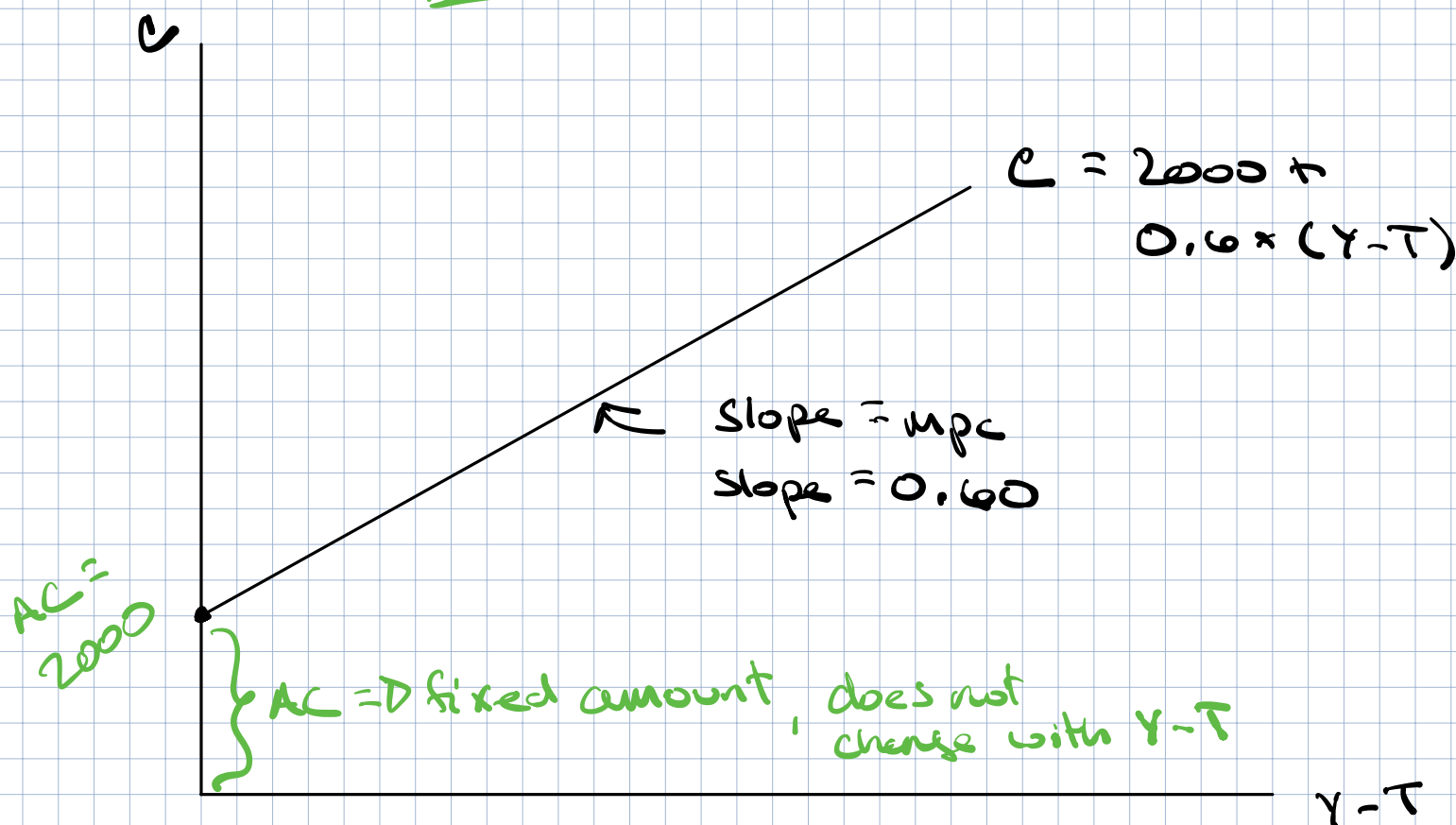
Increase $Y - T$ by \$100 $\Rightarrow$ Increase C by \$60

Increase $Y - T$ by \$1 $\Rightarrow$ Increase C by \$0.60

$$mpc = \frac{\Delta C}{\Delta Y - T} = \frac{1200}{2000} = 0.60$$

$$C = 2000 + 0.6 * (Y - T)$$

Show $C = \underline{2000} + 0.6 \cdot (Y - T)$ visually



Goal: find $Y = AE \rightarrow \text{not } (Y - T) = AE$

Account for taxes (T) $\Rightarrow$ Assume T is fixed

Assume level of $T = 2000$ for all values of Y

<u>Y</u>	<u>T</u> ($T = 2000$)	<u>Y - T</u>	<u>C</u>
2000	2000	0	2000
4000	2000	2000	3200
6000	2000	4000	4400
8000	2000	6000	5600
10000	2000	8000	6800
12000	2000	10000	8000

Convert to equation

Start: $C = AC + mpc \cdot (Y - T)$

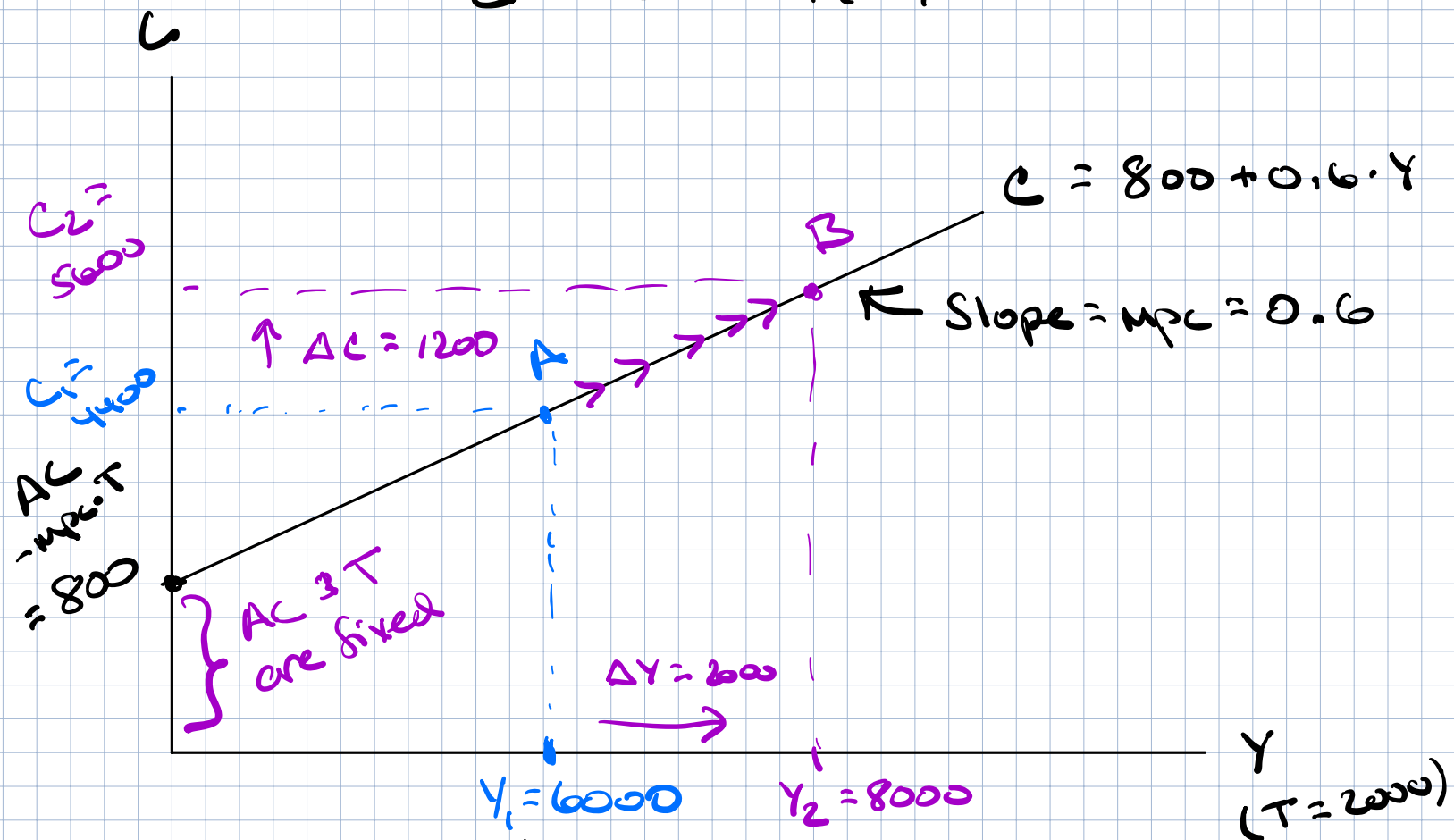
$$C = 2000 + 0.6 \cdot (Y - T) \quad (T = 2000)$$

$$C = 2000 + 0.6 \cdot (Y - 2000)$$

$$C = 2000 + 0.6 \cdot Y - (0.6 \cdot 2000)$$

$$C = (2000 - 1200) + 0.6 \cdot Y$$

$$C = 800 + 0.6 \cdot Y$$



What happens to our consumption function when income changes? *Move along C*

Same consumption function $\rightarrow C$ vs Y

Pt. A $\Rightarrow Y = 6000, T = 2000, Y - T = 4000, C = 4400$

Pt. B $\Rightarrow Y = 8000, T = 2000, Y - T = 6000, C = 5600$

What will change the entire consumption function?

What will change consumption other than income?

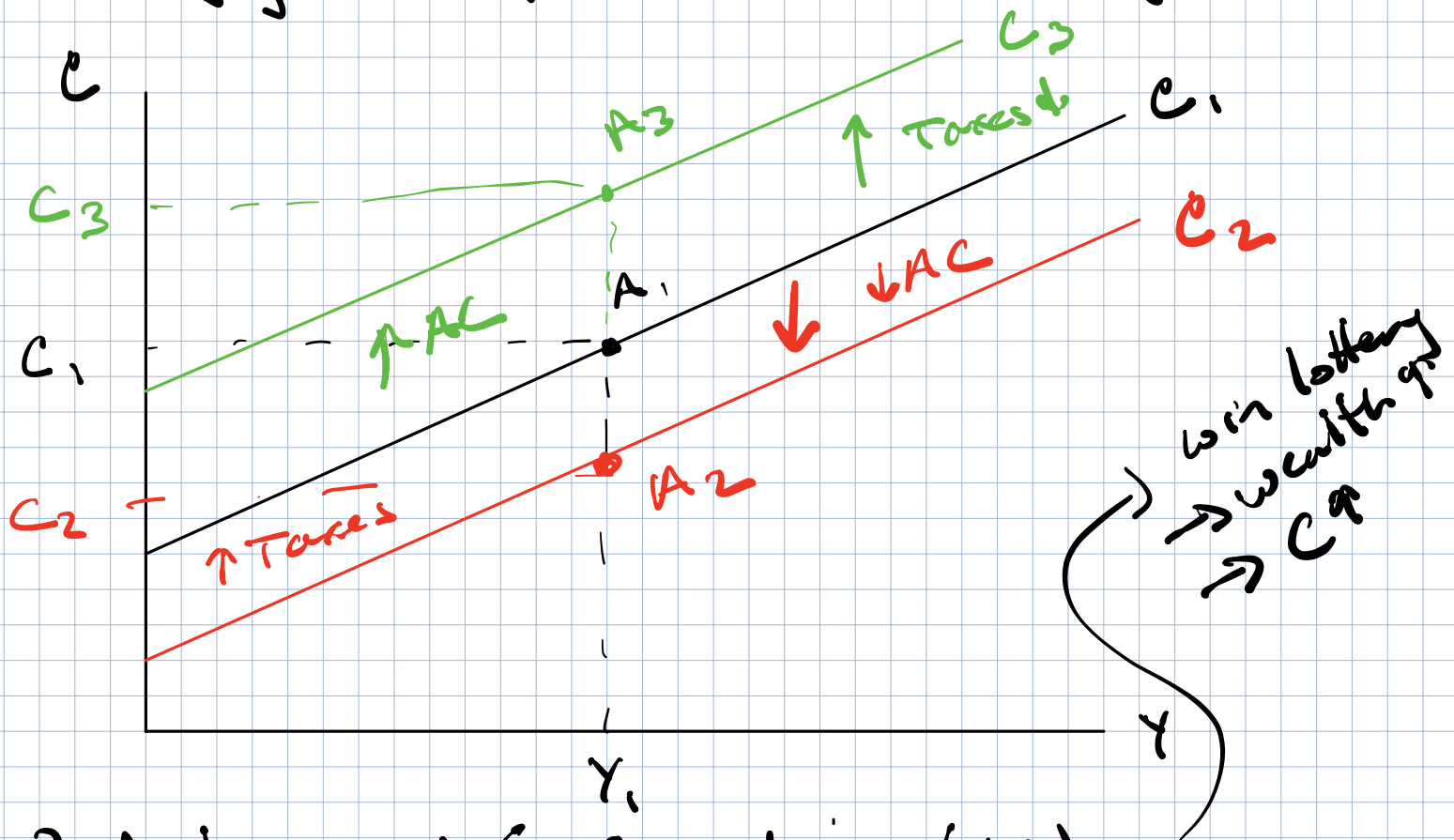
What shifts consumption function?

Two big shifters of $C = AC + mpc \cdot (Y - T)$

1. Taxes (T)

If Taxes $\uparrow \Rightarrow Y - T \downarrow \Rightarrow C \downarrow$
If Taxes $\downarrow \Rightarrow Y - T \uparrow \Rightarrow C \uparrow$ } No change in Y

Changing level of C at all levels of Y



2. Autonomous Consumption (AC)

More kids $\rightarrow$ more C at all levels of Y

Stock market crash $\rightarrow$ wealth $\downarrow \rightarrow C \downarrow$